Chapter 7

Conclusions

7.1 Introduction

This thesis has demonstrated a method of systematic analysis designed to calculate the extent and approximate location of subsistence catchment areas for settlements of the prehistoric southern Levant. The methodology which has been used to model subsistence catchment areas has been based solidly on archaeological data, and further enhanced with the incorporation of the physiological characteristics of the plants and animals found in the palaeobotanical and faunal assemblage as well as by the use of both modern and palaeoenvironmental factors. A systematic analysis using many different combinations of input factors (for example, using a range of values for population, crop yield estimates, contribution of dairy products to the diet and so on) has shown that significant differences in landscape engagement result.

In the introduction to this thesis I proposed several research questions that might provide insight into the phenomena seen in the Early Bronze Age, particularly the appearance of an increased density of large and small settlements in the fourth millennium BC.

> ***If we understand urban communities as (at least partly) requiring access to resources to sustain them beyond those afforded by their immediate hinterland, then insight into the above question of whether these Early Bronze Age communities could be sustained by their immediately adjacent local landscape will surely be a valuable contribution to this debate.***
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> **Questions:**
> ***Did they outgrow the capacity of the immediately adjacent local landscape to support themselves, and what sort of impact on the landscape can we understand such communities to have had?***
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> ***Can an empirically rigorous argument be made that, contrary to the main trend in scholarly literature on the Early Bronze Age over the past 15 years, land-pressures under conditions of population growth in the EBA favoured the development of regional site dependencies, exchange networks and perhaps also settlement hierarchies?***

Two approaches can be adopted to defining immediately adjacent local landscapes.

One approach is traditional within geography and employs concepts of travel time to exploit fields by farming communities. The local landscape is then defined as the land that can be exploited within one days return journey, typically 5 kms, as classically utilised in preliminary site catchment analyses. As a basis for catchment analysis this approach has been seen to be flawed as discussed in Chapter 3.

In particular it has been suggested to impose a modern western economic viewpoint on the landscape. However, in practice many communities that do not operate such ergonomic conceptual frameworks nevertheless operate within such constraints and its is therefore convenient to address this question in reference to such a framework. Using this criterion Shuna, for example, would almost certainly *not* have required crop land beyond these limits and in that sense had *not* outgrown the capacity of the local environment to sustain itself. This approach would conclude that Shuna would most likely not be an appropriate representation of an urban settlement.

A second approach, however, is to understand whether enough land would have been available around sites like Shuna given the demands of the wider settlement world and especially adjacent settlements. In this regard under a number of scenarios there would have been considerable problems for a community like Shuna to meet crop needs from the local landscape and it is plausible other communities grew food made available to at least some at sites like Shuna through exchange or 'taxation'. From this angle, it does seem likely that, given the presence of adjacent settlements as was shown in Chapter 6, the implication is that in this context it could meet the criterion of an urban centre in the sense that it was a community that could not support itself easily on the adjacent available land given the presence of these other settlements. Conversely, if approached from the opposite angle looking first at the smaller settlements, would it have been feasible for them to have had their subsistence catchment area completely engulfed by a much larger settlement *without* this type of relationship?

As stated earlier, the conversion of huge areas of land to cultivation must have occurred to sustain the increasing population. I feel that it has been demonstrated that at a minimum, changes in agricultural strategies had to have occurred to support the populations witnessed in the EBA. At the very least, production levels of food *had* to increase in response to population increases. While clearing of land for use as cultivated cropland was certainly not new, the *scale* at which it must have occurred in the EBA was.

As a result of the regional analysis done in Chapter 6, it is, I believe, quite reasonable to consider the possibility that, as the landscape surrounding settlements became more devoted to the production of cultivated lands (resulting in more distant pastures), these outlying areas could have been places where people involved with the rearing of animals might have moved to on a possibly permanent basis.

Whilst this move may have been one of convenience, it could be interpreted as the beginnings of a form of urbanism. With higher populations came the need for more food. An examination of just one community, as was done with Shuna in Chapter 5, shows that it is possible to sustain any population with any of the scenarios entered into the model. When viewed regionally, however, it has been shown that access to a sufficient area of suitable land becomes a limiting factor. I believe that a better way to approach this question is to ask if communities outgrew the production potential of their subsistence strategy given the area of suitable land to which they had access.

The mere *presence* of smaller sites within what has been shown to even the smallest possible crop land extents (e.g. Figure 6.12 page 339) argues for a more complex asymmetric relationship between these communities that are at the heart of some understandings of urbanism. If considered in the context of higher populations and other modelled scenarios, it becomes even more evident that these relationships must have existed. Although Chesson and Philip argue against urbanism in favour of their corporate village model, I feel that, at least in part, they fail to consider some of these types of relationships that are urban in nature, even if they are not exactly like the form of urbanism we see in Mesopotamia.

Not only has the the scale of the settlement been demonstrated as being an important factor for assessing landscape engagement, but it has also been demonstrated how valuable it is to consider the wider region when assessing the feasibility of proposed land use resulting from various subsistence strategies. The modelling software proved useful in that it has provided a way to quickly and systematically examine so many scenarios. Chapter 6 has indicated that certain areas, such as the Jordan Valley, were likely to have had to have close ties to neighbouring communities, especially as populations continued to climb and the need for cultivated land increased. It has also shown an interesting trend for small sites to be surrounded by the cropland of the larger settlements. Given the coarse temporal resolution available to us, the fact that these small sites *appear* to lie within crop land extents might mean that they were in their communities grazing land extent when first established.

> **Questions:**
> ***Is it possible to find evidence to support the proposition that agricultural intensification must have been associated with the development of the new larger scale of EBA communities?***
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> ***Can we find evidence that, as Philip (2003) and Chesson (2003) put it, such communities 'inscribed' themselves on their landscape in new ways, and are these communities likely to have been some of the first to have had a significant negative impact on their physical environment?***

By examining a large number of different scenarios, it has been shown that certain strategies appear more favourable than others depending in part on circumstance. For example, the suspected 'meatier' diet of the Chalcolithic was observed to be sufficient only up to a certain population level, at which point a change of subsistence strategy appears to be needed if population levels were to continue to climb. Because populations are based on a combined use of site size and population density estimates, it is necessary to talk in terms of population density estimates, as the site sizes are calculated from the archaeological evidence, whilst the human population density can only be estimated. Therefore, given that we have an idea of what the settlement sizes were during the various phases of occupation, maximum and minimum populations should be used. That being said, as time progressed from the EEBI, settlement sizes could be seen to increase, and even using minimum values for population density, population levels corresponding to the site sizes were shown to be problematic under certain subsistence strategies. I suspect that changes in subsistence strategies would have been slow to change until/unless something occurred which made it a necessity. Simply expanding your territory using the same strategy would have worked to a point, but eventually this expansion reaches a limit, forcing something to change. This limit to expansion can be from any number of things, such as natural barriers like river or large bodies of water, changing ecological zones, or competition for land from neighbouring communities. When these limits were reached, a more efficient use of the available land was likely to have had to occur if population levels were seen to have increased. This was shown to be possible throughout the course of Chapter 5, and is supported by the archaeological evidence (e.g. the increasing proportions of cattle perhaps indicating greater ploughing at Shuna, or the 'disappearance' of pigs perhaps being an indication for the intentional drainage of bottom lands, and so on).

With cultivation being a very visible inscription on the landscape already, I propose that other more deliberate attempts to indicate an association of communities to their more distant grazing lands could have been made. Although my research has not been able to explore these issues directly, it seems reasonable that one might find features such as dolmens in these areas, deliberately placed to ensure that the land was in use by them. In addition the packing of landscapes like the Jordan valley with the fields, fallow as well as cultivated, of significant number of adjacent communities seems likely to have demanded systems of the signalling of land ownership more clearly than in the past.

On this basis one suspects visible land boundaries would have been a prominent feature of such densely utilised ares of landscape as the Jordan valley and Jezreel. Likewise long term investment in increasing areas of olive and vine cultivation in the surrounding hills were likely marked by field boundaries and terracing. One is even tempted to imagine the appearance of a classically Mediterranean landscape for the first time in the fourth millennium BC.

As stated above, when a current subsistence strategy had consumed all available suitable land, something had to change, and it has been shown throughout the course of this thesis that this could very well have been a change to the diet/subsistence strategy.

Questions:
How did such relatively large communities sustain themselves within the environments in which they operated?

What sorts of land use can we interpret from the available data on subsistence practices combined with environmental evidence and our understanding of their landscapes?

If we can ascertain the types of land use, what scale of activity relating to the associated land use can we understand to have operated in these circumstances?

After examining a large number of scenarios using set combinations of values for input data that had a range of acceptable values, it has become evident that studies of this nature on settlements in isolation are incomplete; only by examining land use on a regional scale can we begin to understand how subsistence strategies can be linked to environmental evidence, and begin to more fully understand the ancient landscape. Land use requirements suggest extensive arable land was needed for large settlements, often focused largely on alluvial, low lying areas with pasture 'forced' into the hills, but was likely managed around the important 'cash' crops like olive and viticulture ('olive culture' was probably practised by Shuna's inhabitants, as indicated by the significant amounts of olive wood amongst the Shuna charcoals.). My dynamic model for caprine and cattle herd management indicates very significant opportunities for dairying, even without a strong 'dairy' signature in the faunal remains.

It has become clear that as populations grew regionally, so to did the likelihood for increased impact of these communities on their landscapes. Both the specific Shuna case study and the more generalised 'network' analysis indicate that even at minimum likely population levels, very extensive tracts of the Jordan and Jezreel valleys and adjacent hills were under cultivation and subject to regular grazing pressure.

The proposed impacts are supported by the limited and low resolution palynological work from the Huleh (Wilkinson 2003, 26-27) that show an increase in olive pollen in the EBA. At least in certain areas, like in the Jordan valley, the implications are that there almost certainly had to have been a major impact on the shaping of the landscape and on natural processes within the landscape. Given the limited scope of analysis possible in this thesis, statements like this remain general on the whole, but certainly, for this specific area, this is a precise consequence of the diets, subsistence strategies and populations that have being modelled.

Because it has been shown that there was this high likelihood for such intensive use of and modification to the landscape, future development of this work will benefit from the contributions of studies dealing with some wider issues that have been raised about soil erosion, land nutrient depletion, deforestation, and so on (e.g. Wilkinson et al. 2007, Wilkinson 2003, Henkin et al. 2005, Blumler 1993, Fall et al. 2002, Seligman 1996).

Even a quick, casual glance at the outcome of the regional analysis done in Chapter 6 makes it obvious that these communities must have had a major impact, for example, on the natural vegetation of the area. Grazing would have affected vegetation, the woodland in the hills would have been affected by the clearance, and the wetland areas in the valley must have also been affected. Even when looking at scenarios which indicated the smallest areas of required cropland, on a regional scale this would have required huge amounts of land to be cleared and put under cultivation. These unprecedented changes to the landscape must have presented these ancient communities with many new problems with which they would have had to learn to deal with for the first time.

Certainly, given the scale of cultivation alone, there would have been potential for massive impact on the land. If people responded to any of the new challenges resulting from these changes in the wrong way, there most certainly is the possibility that it could have had very negative effects on their surrounding environment.

> ***The use of site catchment analysis helps address several of these questions, but previous 'site catchment' studies have generally only focussed on individual settlements; however, settlements and communities exist in networks. In particular, one of the notable features of the Early Bronze settlement record is the obvious existence of networks of communities, especially in the larger communities (Philip 2001, 207).***
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> **Question:**
> ***What can a combined study of land use and routes connecting a network of settlements allow us to understand about relationships between settlements and the nature of such networks?***

The development of Landuse Analyst has proven to be a very efficient and useful way utilisation as many input factors as possible into a land use model. Although it took a huge amount of time to design and create, the payoff has been worthwhile; it is now possible to very quickly explore any conceivable number of combinations of factors in a very short amount of time, and provides a tool which will provide a systematic way of comparing results across a wide range of archaeological scenarios.

Although somewhat preliminary in nature, Chapter Six has shown that a Landuse Analyst can effectively be employed together with other GIS techniques to offer insights into landscape use not possible in the past. The analysis done in this research really only scratches the surface of insights which this kind of work can yield. With further sophistication to (and improved automation) of site catchment analysis, and faster hardware that will allow increasing resolutions of data, the combination of settlement land use modelling and regional network analysis will prove exciting indeed.

7.2 Methodological Development and Future Work

Developing a methodology for use in this thesis resulted in the creation of Landuse Analyst, a tool which has the capacity to provide a large number of different avenues for future research.

There is a wealth of information provided by Landuse Analyst that I have not been able to touch upon in this thesis. For example, as a result of the herd re-constructions, the number of adult females, males, newborns and juveniles is calculated. These estimates could be used to provide further insights into the relationships between the animals and the inhabitants of the settlement. For example (using conceptual numbers), if the settlement being modelled had a population of 3000 people and a total cattle herd of 750 adults, it could suggest that every 4 people had one cow, or if one were to estimate a household family size at eight persons, 2 cows per family. One could also approach this from other angles, such as using ethnographic studies to determine how many cows one shepherd (or a family unit) could manage. If that number was 15 (again, conceptual numbers for illustrative purposes only), then 50 shepherds (or 50 family units) could have looked after the cattle, leaving 2600 people (or 325 families) to deal with the goats, sheep, crops, pot making, and so forth. Reconstructing animal management strategies like this could allow for a more sophisticated analysis of landuse, whereby one might examine on a local level strategies which could have been employed for the sharing of grazing land. This could be put into a more temporal perspective as well, taking into account when animals would have been wanted for use in the fields as draught animals, or kept nearby to maximise the collection of milk for the production of dairy products, in conjunction with the annual climatic cycle and the various ecological zones available to them.

Increasing importance of route location has been linked to viticulture intensification (Gibson 2006, 78), which would be an interesting avenue to explore in a higher resolution analysis. One of the issues raised is that leaking liquids or crushed fruit from vine crops result in slippery trails, so careful clearing of routes was done as a way of safeguarding the transport of this important produce. Even at this lower resolution, it is interesting to see that the routes through hill-country, which is also the landscape which is best suited for vine crops, tend to follow the wadi bottoms. This might be problematic for transport of liquids, as vegetation can be dense in these areas. Routes through the wadis therefore might have been cleared, and their constant use would have resulted in hard packed soils. This, and the lack of vegetation on these trails could have accelerated erosion, and resulted in the need for a constant changing of route locations over time. This may have, at times, encouraged trade or exchange with more distant communities which had less risky routes; likewise, it could also have just resulted in a more difficult route to be used. To better investigate this type of route placement in future studies, it would be beneficial to closely look at ground cover in addition to topographical features to allow these factors to be taken into consideration.

Surplus potential can also be examined. Landuse Analyst can limit the amount of dairy which contributes to the diet, and calculates any surplus. This data could be used in conjunction with regional network studies exploring potential trade relations between settlements, or to see if some settlements were better suited physiographically to focus on the overproduction of certain commodities to keep other settlements (which may have been less suited for the production of this product) supplied.

Landuse Analyst also calculates animal density, and could prove especially insightful when looking at animals which could have provided traction. For example, is there any evident change in the number of adult cattle when the area of cultivated land being used increases? If there is, does it coincide with the use of the plough? If so, did this in turn make the production of certain crops for use as fodder more feasible, thus lowering the land requirements for grazing? As pointed out in Chapter 5, these relationships are supported by the archaeological record at Shuna (Chapter 5, page 313). This recursive relationship with the archaeological record means that it will be possible to address some of the new questions arising from this type of analysis.

By changing the contributions of non-domestic food items in Landuse Analyst, it is also possible to examine scenarios where people would have resorted to an increased utilisation of these resources (e.g. fishing, hunting, collecting of other wild plants) during times of stress (like drought or crop failure due to disease).

Clearly, the resolution of the geographical data used to classify land suitable for use by the crops and animal herds can always be improved. The simple approach taken in this thesis was to use only one classification factor (slope), but Landuse Analyst has been specifically designed to use any combination of factors. Factors such as soil types, rainfall patterns, more complex crop rotations, salinity problems and so on can be taken into consideration when setting up the model.

Crops and animals can, on an individual basis, be given either sole access to, or the first choice to access specific land, as well as to land deemed suitable for use by all crops. Grazing of fallow land can be done in a more sophisticated manner as well. In Chapter 5, where detailed analysis was done at Shuna, all animals were given equal access to fallow land for grazing.

Landuse Analyst does provide an easy system for granting access to fallow land for grazing to the animals using a three level access priority system, so that if one was wishing to look at the use of grazing lands in more detail, this could be used to do do so in a very detailed way. For example, if cattle were given first priority to all fallow land in order to keep them close to the settlement for dairy production and traction purposes, this might change the herding strategies used further out from the settlement.

The archaeological record can only provide us with clues as to what the crops being grown were. By looking at remains which might represent several hundred years, annual variations of sowing choices are impossible to know. Landuse Analyst provides an easy way to explore if there is an effect from sowing crops in varying ratios. By simply changing the contribution levels of an individual crop (or animal for that matter) one can instantly see the results. If, for example, one wanted to see what might have happened if people responded to dry conditions at the time of sowing by planting more barley, this can easily be done. All of the potential effects, including foddering animals, dairy production contributions, fallow grazing, etc, are instantly calculated. The ability to explore these types of temporally high resolution scenarios could provide interesting insights into how ancient peoples responded to different events especially as the resolution of the palaeoenvironmental record continues to improve as does the modelling of palaeoenvironments.

It also allows for an examination of whether temporary increases in production of certain items could have been possible. For example, if one wanted to explore the situation where one settlement experienced a temporary food shortage due to localised drought for example, it could be modelled by changing population levels. This would be done by first noting the levels required and produced by the drought stricken settlement, and then cutting their population level according to the severity of the drought; if for example, you wanted to simulate a 75% failure, and the settlement had a population of 1000 people, one could cut the population by 75%, and note the production levels. This shortage could be added onto a neighbouring site by adding the population cut from the drought stricken site to it's own. By manipulating crop rotation values or the ratios of the crops being sown, it is possible to look for a way in which this extra production might have been incorporated into the agricultural land at the disposal of the site providing the assistance. The effects of interrupting their normal cycle of crop rotation could even be explored for the following year by lowering yields of the crops to simulate the negative effect of sowing the fallow land to help out the neighbours.

By manipulating the values used to define the animals in Landuse Analyst, it is possible to simulate the use breeding stock in animal herds for emergency food. For example, one could increase the number of offspring per birthing event, while at the same time lowering the size of the offspring at slaughter. To explain, it is necessary to build a small scenario. Assume that an animal normally gives birth one baby a year, and the weight at slaughter is 100 kg. If you wish to model the halving of the breeding herd, you can give that animal two babies per year, and reduce the slaughter weight to 50 kg. This will effectively halve the breeding herd size (thus halving the potential dairy production levels as well). Further tweaks are possible as well, such as adjustments to animal food requirements and so on that might be used to provide increasingly sophisticated scenarios that might be wanted in order to simulate changes to foddering strategies and so forth.

7.3 Concluding Remarks

I have demonstrated in this thesis a method of using palaeobotanical and faunal assemblages to help reconstruct potential subsistence strategies and land use models, and provided a detailed implementation of this methodology at the south Levantine site of Tel esh Shuna North in the Jordan Valley. The results of this analysis were further extended to incorporate 125 additional sites, providing a regional context which allowed further observations and isights to be made. By focussing on the inclusion of as much archaeological evidence as possible, the reconstructed environments which emerge from this analysis avoid many of the problems associated with more theoretical approaches. The 'bottom-up' approach taken in this research allows for a closer relationship to the archaeological material, facilitating the synthesis of diet, subsistence strategies and land management techniques between many forms of archaeological evidence in a clear, systematic and replicable manner. By creating Landuse Analyst, (an open source tool that is freely available to anyone, runs on any computer, and which can be used for any time period at any location in the world), I have not only moved beyond the usual restraints imposed by the use of proprietary software packages (e.g. cost, data format, operating system requirements, and so on), but I have also avoided the problem of developing a methodology specific to, and directly applicable to, only the area of study in the topic of this research thus allowing future researchers to directly utilise these methods (via Landuse Analyst) into their own unique and specific areas of interest.

In the introduction to this thesis I highlighted several questions related to settlement patterns of the early bronze age. Using the site of Shuna I have illustrated not only how land use can vary depending upon the particular characteristics of a subsistence strategy according to variations in the diet, but also how these strategies would likely have had to change over time to accommodate an increasing population. As communities grow larger, the subsistence strategies, which are constrained to the environment in which they operated, had to change when/if this environment became fully exploited. The size of a prehistoric settlement would also have affected its impact upon the surrounding environment and resultantly the strategies used by people to modify the landscape to suit their needs.

It has been shown in this thesis that by the EEBI, the southern Levant could have indeed been confronted with this issue. If we look at the maximum (or even average) scenarios, it actually seems quite likely that already by the EEBI these communities would have had a pretty significant impact; the majority of the Jordan Valley has been shown to have likely been nearly completely devoted to use as croplands, and it is likely that significant areas adjacent to the valley would have been taken into fields and terraces, coupled with large amounts of land which would have had to have been in use as pasture. This all suggests that it seems extremely likely that these communities for the first time had significantly altered their landscape and impacted their environment, and thus, that they did in no small way inscribe themselves on their landscape.